

Assignment 1

Exploratory Data Analysis (EDA)

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Chicago Food Inspections

Chicago Food Inspections

From City of Chicago Open Data



Dataset file:

https://drive.google.com/file/d/1Viw5f1rVIhtRWWhMHOLAvbDSOHiXEwVT6/view?usp=drive_link

Dataset Description:

Dataset Source:

Chicago Food Inspections dataset obtained **from Kaggle**.

<https://www.kaggle.com/datasets/chicago/chicago-food-inspections/data>

Number of Rows and Columns:

The dataset contains 196,825 rows and 22 columns before cleaning.

```
No. of rows and columns

# finding number of rows and columns
print("Shape (rows, columns): ", df.shape, "\n")

print("number of rows: ", df.shape[0])
print("number of columns: ", df.shape[1])

[74] ✓ 0.0s Python

... Shape (rows, columns): (196825, 22)

number of rows: 196825
number of columns: 22

Generate Code Markdown
```

The dataset contains 196676 rows and 15 columns after cleaning.

```
No. of rows and columns after cleaning

> print("Shape (rows, columns) after cleaning: ", df.shape, "\n")
print("number of rows: ", df.shape[0])
print("number of columns: ", df.shape[1])
[38] ✓ 0.0s Python

... Shape (rows, columns) after cleaning: (196676, 15)

number of rows: 196676
number of columns: 15
```

Description of Features (Columns):

- Inspection ID: **Unique identifier for each inspection.**
- DBA Name: **Official name of the business.**
- AKA Name: **Alternative name of the business.**
- License #: **Business license number.**
- Facility Type: **Type of facility, such as restaurant or school.**
- Risk: **Risk category assigned to the facility.**
- Address: **Street address of the business.**
- City: **City where the business is located.**
- State: **State of the business location.**
- Zip: **Zip code of the business location.**

- Inspection Date: **Date when the inspection was conducted.**
- Inspection Type: **Type of inspection performed.**
- Results: **Outcome of the inspection, such as Pass or Fail.**
- Violations: **Reported violations found during the inspection.**
- Year: **Year extracted from the inspection date for time-based analysis.**

Purpose of Using this Dataset:

The purpose of using this dataset is to analyze the results of food inspections, identify patterns in inspection results, and study trends over time to better understand compliance levels and food safety performance. Therefore, this dataset was used to determine which restaurants passed inspections and which did not.

data cleaning:

Data Cleaning Steps:

The following data cleaning steps were performed to prepare the dataset for analysis:

- Removed missing values from important columns.
- Checked and removed duplicate records.
- Converted data types (such as dates) into the correct format.
- Standardized column names and values for consistency.
- Extracted useful features such as the year from the inspection date.

Why Cleaning Was Needed:

Data cleaning was necessary to ensure the quality and accuracy of the analysis.

The dataset contained issues such as:

Missing values in some columns

Duplicate records

Inconsistent data formats (especially dates)

Irregular or unclear values in some fields

These issues could affect the results and lead to incorrect insights if not handled properly.

Before cleaning:

	Inspection ID	DBA Name	AKA Name	License #	Facility Type	Risk	Address	City	State	Zip	...	Results	Violations	Latitude	Longitude
0	2352683	CHIPOTLE MEXICAN GRILL	CHIPOTLE MEXICAN GRILL	2670642.0	Restaurant	Risk 1 (High)	1025 W ADDISON ST	CHICAGO	IL	60613.0	...	Pass w/ Conditions	3. MANAGEMENT, FOOD EMPLOYEE AND CONDITIONAL E...	41.947118	-87.655381
1	2352734	CHILI'S T-I	CHILI'S (T1-B14)	34169.0	Restaurant	Risk 1 (High)	11601 W TOUHY AVE	CHICAGO	IL	60666.0	...	Pass	10. ADEQUATE HANDWASHING SINKS PROPERLY SUPPLI...	42.008536	-87.914428
2	2352713	NICK'S FIRE GRILL STEAK & LEMONADE INC.	NICK'S FIRE GRILL STEAK & LEMONADE INC.	2699090.0	NaN	Risk 1 (High)	2900 W 63RD ST	CHICAGO	IL	60629.0	...	Not Ready	NaN	41.779190	-87.695892
3	2352701	TAQUERIA BLUE LINE INC.	NaN	2703436.0	Restaurant	Risk 1 (High)	3401 W BELMONT AVE	CHICAGO	IL	60618.0	...	Pass w/ Conditions	3. MANAGEMENT, FOOD EMPLOYEE AND CONDITIONAL E...	41.939168	-87.712546
4	2352727	PORTAGE PARK DAY NURSERY	MOSAIC EARLY CHILDHOOD ACADEMY	2215815.0	Children's Services Facility	Risk 1 (High)	5332-5334 W ADDISON ST	CHICAGO	IL	60641.0	...	Pass	NaN	41.946065	-87.760722
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
196819	67757	DUNKIN DONUTS/BASKIN-ROBBINS	DUNKIN DONUTS/BASKIN-ROBBINS	1380279.0	Restaurant	Risk 2 (Medium)	100 W RANDOLPH ST	CHICAGO	IL	60601.0	...	Pass	NaN	41.884586	-87.631010
196820	52234	Cafe 608	Cafe 608	2013328.0	Restaurant	Risk 1 (High)	608 W BARRY AVE	CHICAGO	IL	60657.0	...	Pass	NaN	41.938007	-87.644755
196821	104236	TEMPO CAFE	TEMPO CAFE	80916.0	Restaurant	Risk 1 (High)	6 E CHESTNUT	CHICAGO	IL	60611.0	...	Fail	18. NO EVIDENCE OF RODENT OR	41.898431	-87.628009

● Update is ready, click to restart.

Ln 2, Col 1 (38 selected) Spaces: 4 Spaces: 4 LF 65 Cell 19 of 50

Facility Type	4760
Risk	69
Address	0
City	139
State	42
Zip	50
Inspection Date	0
Inspection Type	1
Results	0
Violations	52266
Latitude	690
Longitude	690
Location	690
Historical Wards 2003-2015	196825
Zip Codes	196825
Community Areas	196825
Census Tracts	196825
Wards	196825
dtype: int64	

Check duplicate rows

```
# checking duplicate rows
df.duplicated()[df.duplicated()==True]
```

[17] ✓ 0.6s Python

```
... 10001 True
    10002 True
    10003 True
    10004 True
    10005 True
    ...
    90059 True
    90062 True
    90063 True
    90064 True
    90066 True
Length: 149, dtype: bool
```



After cleaning:

	Inspection ID	DBA Name	AKA Name	License #	Facility Type	Risk	Address	City	State	Zip	Inspection Date	Inspection Type	Results	Violations	Yea	
	0	2352683	CHIPOTLE MEXICAN GRILL	CHIPOTLE MEXICAN GRILL	2670642.0	Restaurant	Risk 1 (High)	1025 W ADDISON ST	CHICAGO	IL	60613.0	2019-12-04	License Re-Inspection	Pass w/ Conditions	3. MANAGEMENT, FOOD EMPLOYEE AND CONDITIONAL E...	2019
	1	2352734	CHILI'S T-1	CHILI'S (T1-B14)	34169.0	Restaurant	Risk 1 (High)	11601 W TOUHY AVE	CHICAGO	IL	60666.0	2019-12-04	Canvass	Pass	10. ADEQUATE HANDWASHING SINKS PROPERLY SUPPLI...	2019
	2	2352713	NICK'S FIRE GRILL STEAK & LEMONADE INC.	NICK'S FIRE GRILL STEAK & LEMONADE INC.	2699090.0	Unknown	Risk 1 (High)	2900 W 63RD ST	CHICAGO	IL	60629.0	2019-12-04	License	Not Ready	No Violations	2019
	3	2352701	TAQUERIA BLUE LINE INC.	Not Available	2703436.0	Restaurant	Risk 1 (High)	3401 W BELMONT AVE	CHICAGO	IL	60618.0	2019-12-04	License	Pass w/ Conditions	3. MANAGEMENT, FOOD EMPLOYEE AND CONDITIONAL E...	2019
	4	2352727	PORTAGE PARK DAY NURSERY	MOSAIC EARLY CHILDHOOD ACADEMY	2215815.0	Children's Services Facility	Risk 1 (High)	5332-5334 W ADDISON ST	CHICAGO	IL	60641.0	2019-12-04	Canvass	Pass	No Violations	2019

```

Inspection ID    0
DBA Name         0
AKA Name         0
License #        0
Facility Type    0
Risk             0
Address          0
City             0
State           0
Zip              0
Inspection Date  0
Inspection Type  0
Results          0
Violations       0
Year             0
dtype: int64

```

duplicate rows

```

7) df = df.drop_duplicates()
✓ 1.0s Python

8) df.duplicated()[df.duplicated()==True]
✓ 1.1s Python

Series([], dtype: bool)

```

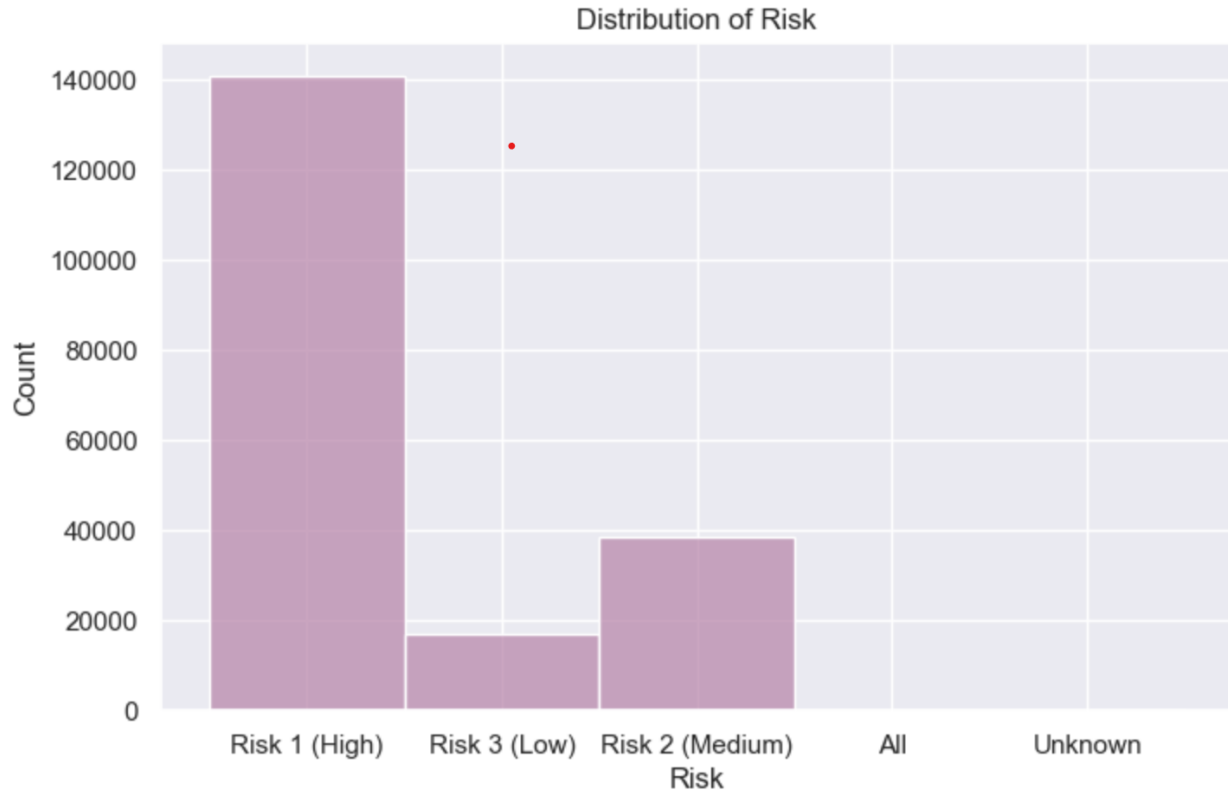


Explanation of Changes:

After cleaning the dataset:

- The data became more structured and consistent.
- Missing and duplicate values were handled.
- Data types were corrected.
- The dataset became ready for accurate analysis and visualization.

visualization, and insights:

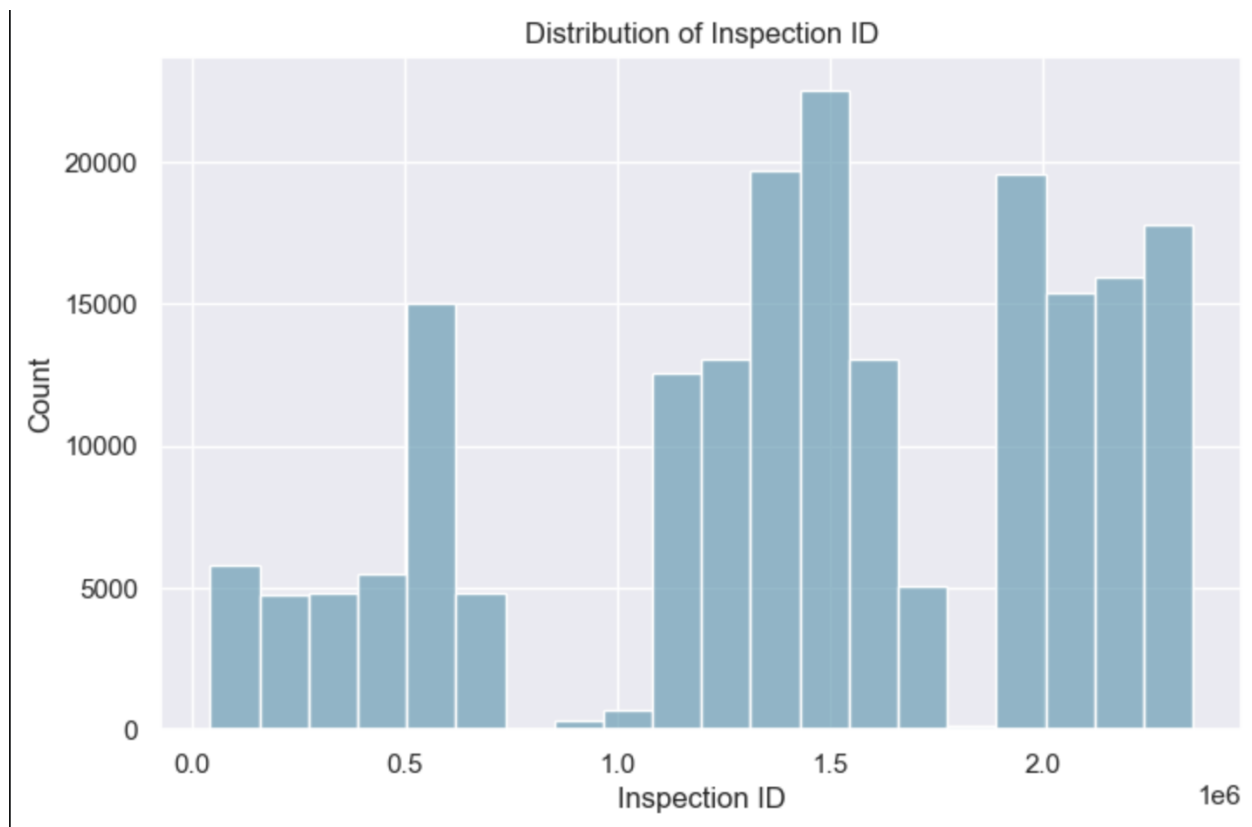


This visualization was chosen to clearly present the distribution of risk levels in the dataset, as a histogram/bar chart makes it easy to compare the frequency of each category.

The main insight shown in the chart is that "Risk 1 (High)" has the highest number of inspections, followed by "Risk 2 (Medium)", while "Risk 3 (Low)" appears less frequently.

An important pattern observed is the imbalance between risk levels, where high-risk facilities dominate the dataset. This comparison highlights that inspections are not evenly distributed across all categories.

From these results, we can learn that inspection efforts are more focused on high-risk establishments, which suggests that these facilities require more attention due to potential safety concerns.

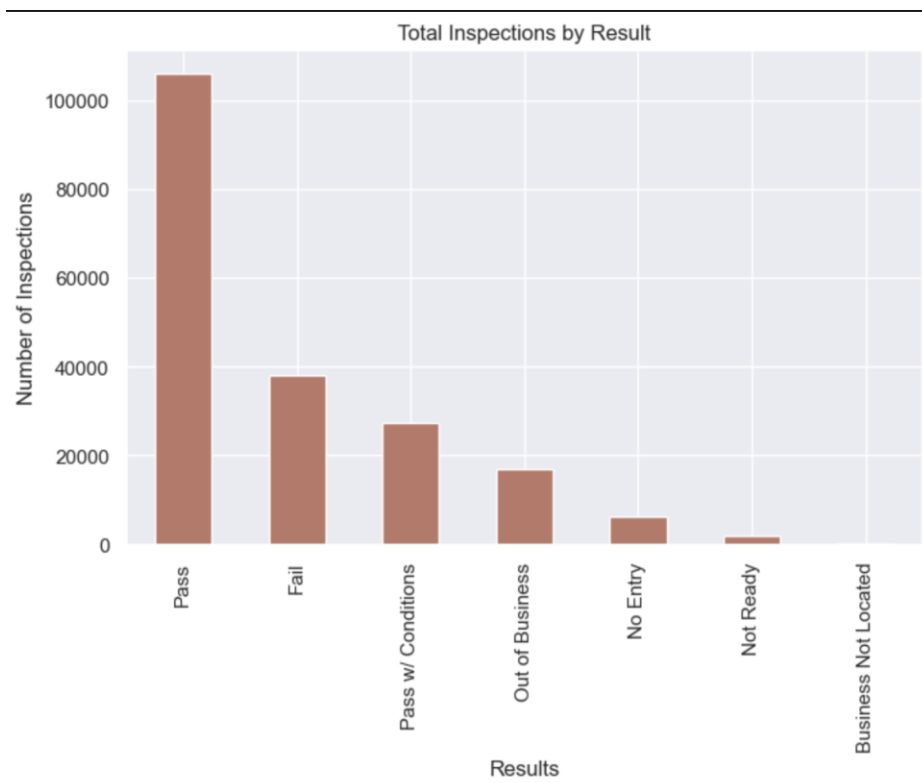


This visualization was chosen to understand the distribution of Inspection ID values, as a histogram helps in identifying how the values are spread and whether there are any patterns or irregularities.

The main insight shown in the chart is that the Inspection IDs are spread across a wide range, with higher concentrations in certain intervals, indicating that inspections were recorded in batches or periods.

An important pattern observed is that the distribution is not uniform, and there are noticeable clusters where many IDs fall within similar ranges. This suggests that inspection records increased during specific periods.

From these results, we can learn that the Inspection ID itself does not carry meaningful analytical value beyond identification, but the distribution indicates possible trends in data collection over time.

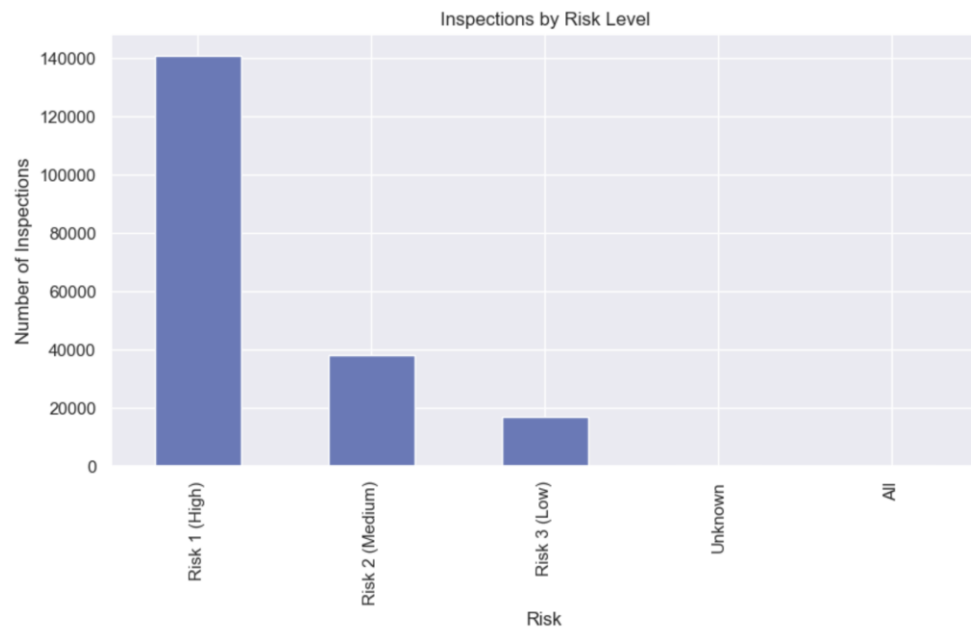


This visualization was chosen to compare the number of inspections across different result categories, as a bar chart is effective for showing differences between groups.

The main insight shown in the chart is that the majority of inspections result in "Pass", while "Fail" and "Pass with Conditions" occur less frequently. Other outcomes such as "No Entry" and "Not Ready" appear in much smaller numbers.

An important pattern observed is the significant imbalance between categories, where successful inspections dominate the dataset. This comparison highlights that passing inspections is far more common than failing ones.

From these results, we can learn that most establishments comply with health regulations, while only a smaller portion fails inspections or requires corrective actions.

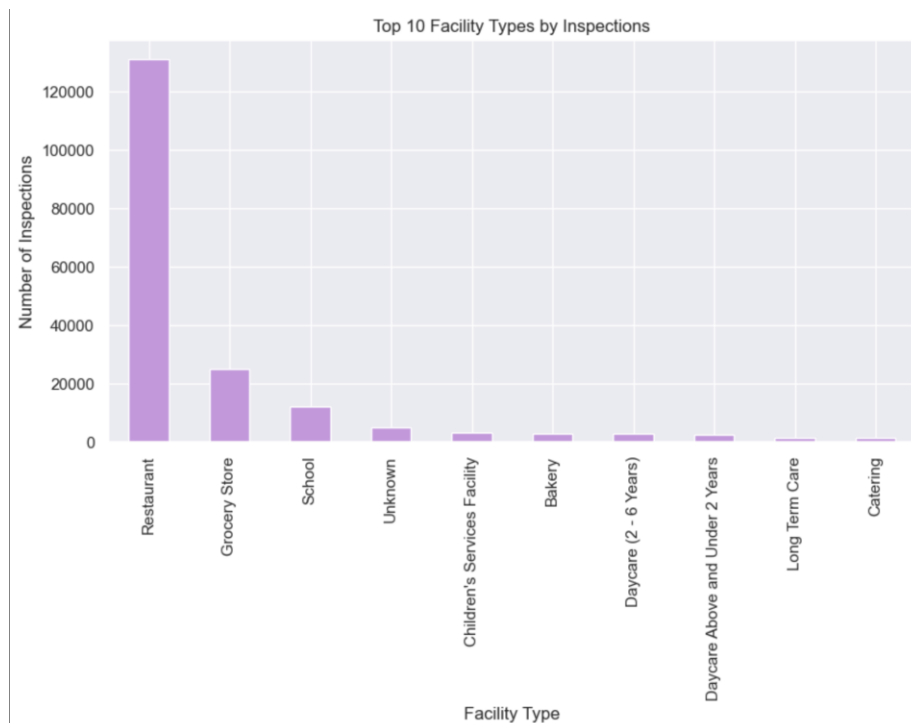


This visualization was chosen to compare the number of inspections across different risk levels, as a bar chart clearly highlights differences between categories.

The main insight shown in the chart is that "Risk 1 (High)" has the highest number of inspections, followed by "Risk 2 (Medium)", while "Risk 3 (Low)" has the lowest count.

An important pattern observed is the strong imbalance between risk levels, where high-risk facilities dominate the dataset. This comparison indicates that inspections are heavily concentrated on higher-risk establishments.

From these results, we can learn that regulatory efforts are primarily focused on high-risk facilities, likely because they pose greater potential health and safety concerns.

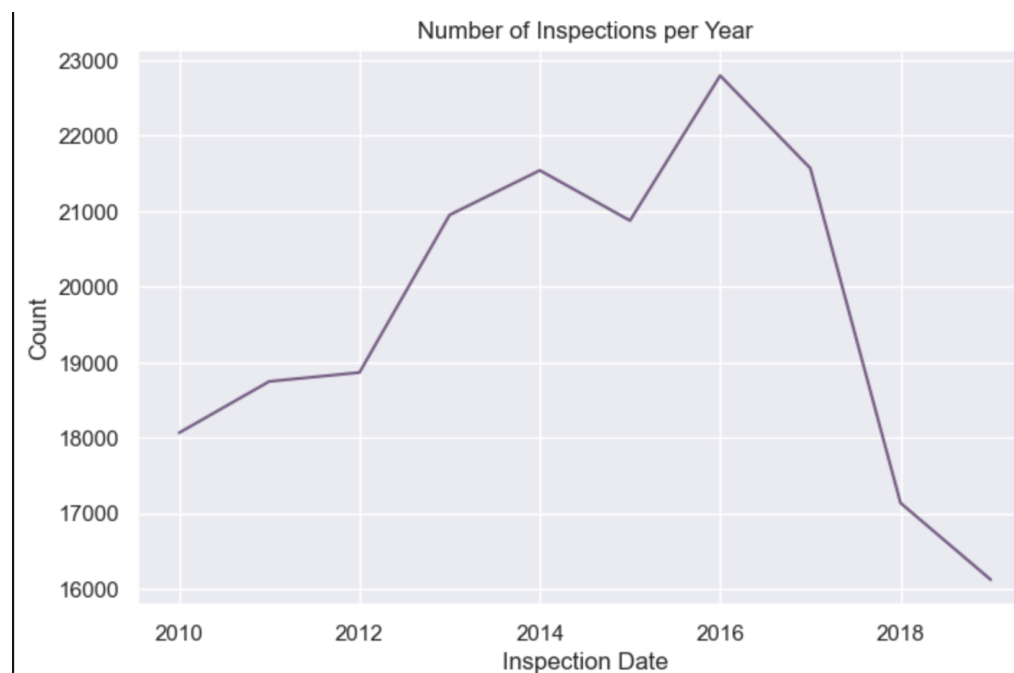


This visualization was chosen to compare the number of inspections across different facility types, as a bar chart is effective for highlighting differences between categories.

The main insight shown in the chart is that restaurants have the highest number of inspections by a large margin, followed by grocery stores and schools, while other facility types appear much less frequently.

An important pattern observed is the significant dominance of restaurants compared to other categories, creating a strong imbalance. This comparison shows that certain facility types are inspected far more often than others.

From these results, we can learn that inspection efforts are primarily focused on restaurants, likely because they represent the largest portion of food-related businesses and may pose higher public health risks.

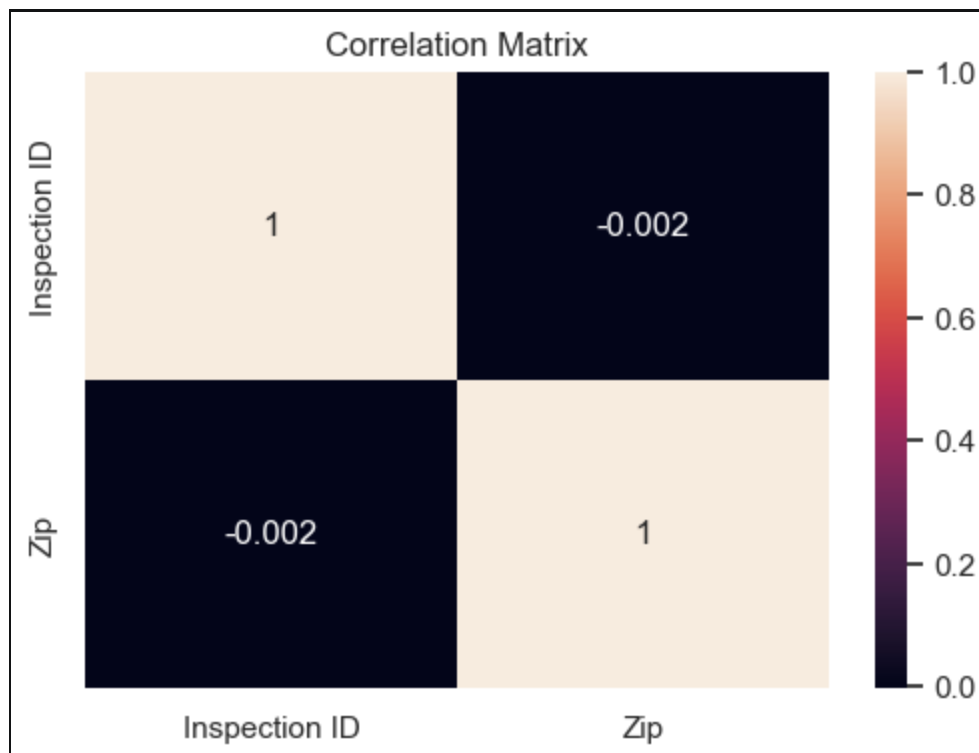


This visualization was chosen to analyze how the number of inspections changes over time, as a line chart is ideal for identifying trends across years.

The main insight shown in the chart is that the number of inspections generally increased from 2010 to around 2016, where it reached its peak, followed by a noticeable decline in the following years.

An important trend observed is the upward growth in inspections during the earlier years, followed by a sharp decrease after 2017. This pattern indicates that inspection activity was not consistent over time.

From these results, we can learn that inspection frequency varies across years, which may be influenced by external factors such as policy changes, resource availability, or operational decisions.

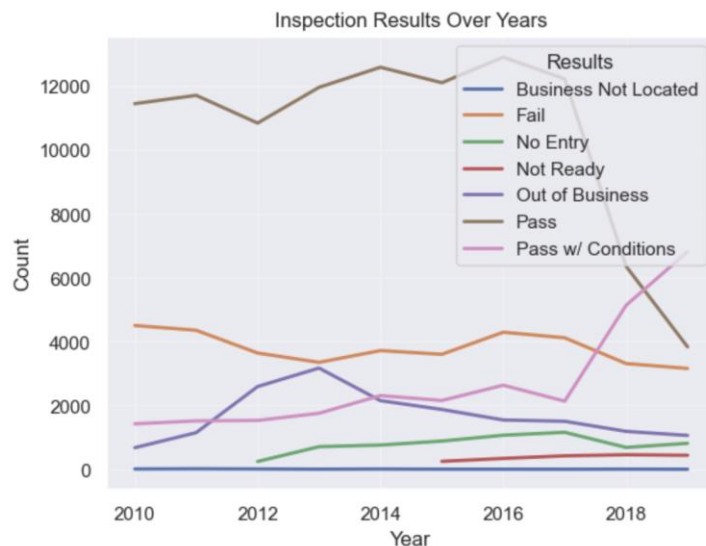


This visualization was chosen to examine the relationship between numerical variables using a correlation matrix, as it helps identify whether there is any linear association between them.

The main insight shown in the chart is that there is almost no correlation between "Inspection ID" and "Zip", with a value close to zero (-0.002).

An important pattern observed is the absence of any meaningful relationship between the variables, as the correlation value is extremely low and close to zero. This indicates that changes in one variable do not affect the other.

From these results, we can learn that these variables are independent of each other and do not provide useful insights when analyzed together, especially since "Inspection ID" is just an identifier and does not carry analytical significance.



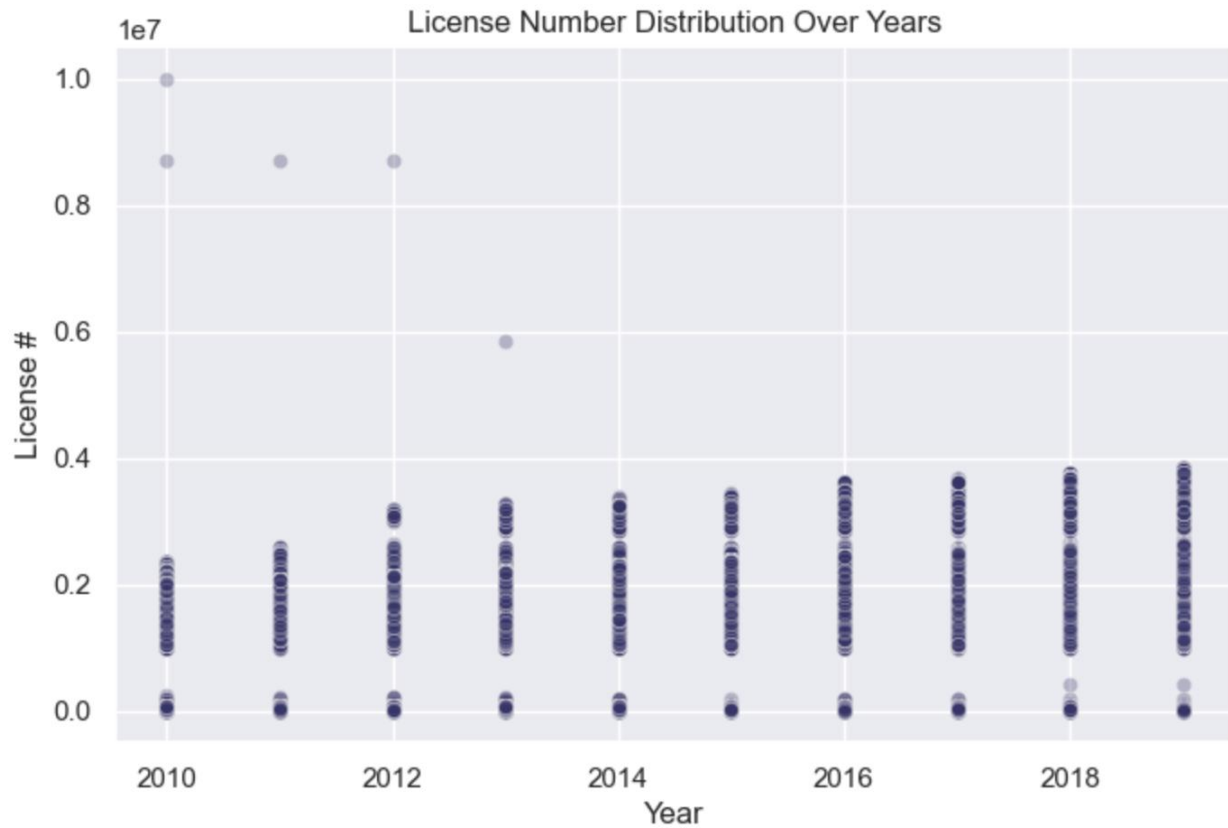
This visualization was chosen to analyze inspection results over time, as it helps identify trends and patterns across different years.

The main insight shown in the chart is that "Pass" results consistently had the highest counts in earlier years, indicating that most inspections were successful. However, there is a noticeable decline in "Pass" results toward the later years.

An important pattern observed is the significant increase in "Pass w/ Conditions" in recent years, especially around 2018 and 2019. This suggests that while businesses are still passing inspections, more of them are doing so with conditions rather than full compliance.

Additionally, "Fail" results show a slight decrease over time, while categories like "No Entry" and "Not Ready" remain relatively low and stable.

From these results, we can conclude that inspection standards may have become stricter over time, leading to fewer full passes and more conditional passes. This indicates a shift toward more regulated and controlled inspection outcomes.



This visualization was chosen to examine how license numbers are distributed across different years using a scatter plot, which helps reveal patterns, variability, and potential outliers.

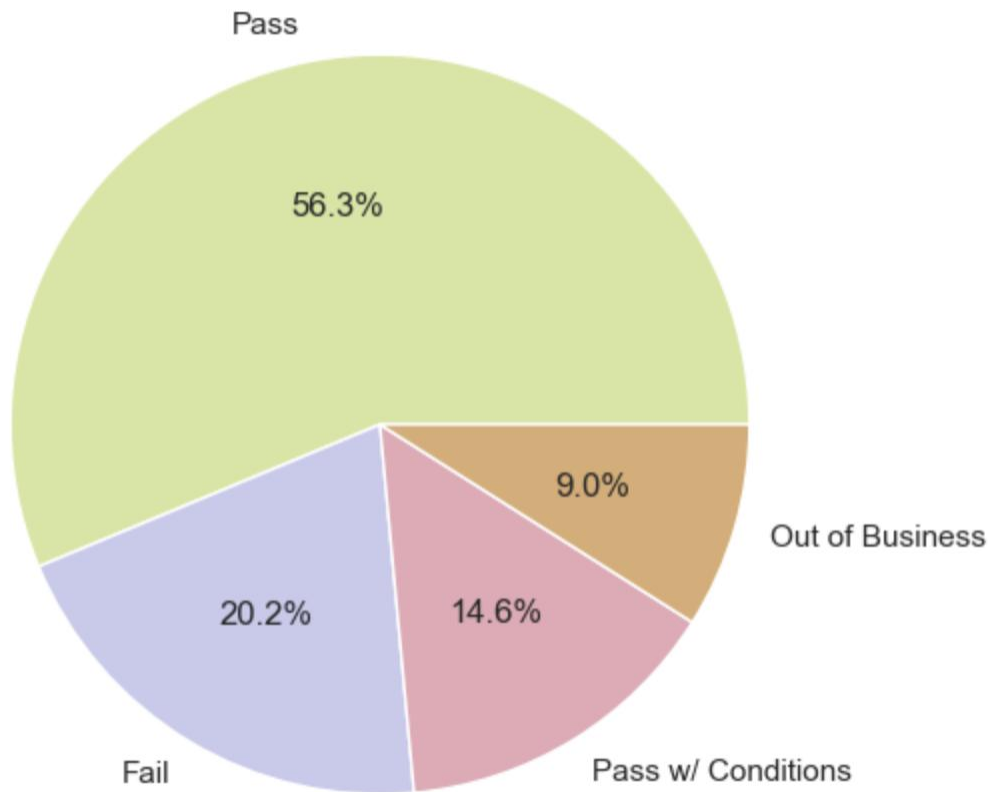
The main insight shown in the chart is that license numbers are widely distributed across all years, with no clear increasing or decreasing trend over time.

An important pattern observed is the clustering of most data points within a specific range, while a few points appear significantly higher than others. These represent outliers with unusually large license numbers.

Additionally, the vertical spread of points within each year indicates high variability in license numbers, suggesting that businesses registered in the same year can have very different license values.

From these results, we can conclude that license numbers do not have a meaningful relationship with time and mainly function as identifiers rather than analytical variables.

Inspection Results Distribution



This visualization was chosen to show the overall distribution of inspection results using a pie chart, as it provides a clear comparison of proportions between different categories.

The main insight from the chart is that the majority of inspections resulted in "Pass", accounting for 56.3% of the total. This indicates that more than half of the businesses meet the required standards.

An important observation is that "Fail" represents 20.2% of the results, which is a significant portion and suggests that a notable number of businesses do not meet inspection requirements.

Additionally, "Pass w/ Conditions" accounts for 14.6%, indicating that some businesses meet standards but still require improvements. The "Out of Business" category represents 9.0%, showing that a smaller portion of businesses are no longer operating.

From these results, we can conclude that while most inspections are successful, there is still a considerable percentage of businesses that either fail or require corrective actions. This highlights the importance of continuous monitoring and regulatory enforcement.